

Weak normalization of the simply typed λ -calculus

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In this note, we review a weak normalization proof of the simply typed λ -calculus based on a decreasing measure for terms based on multisets.

Definition 1 (Simply typed λ calculus, à la Church). The set of types is given by:

$$A ::= \alpha \mid A \rightarrow A$$

Typing rules are given by:

$$\frac{}{\vdash x^A : A} \text{ax} \quad \frac{\vdash t : B}{\vdash \lambda x^A. t : A \rightarrow B} \rightarrow \text{I} \quad \frac{\vdash t : A \rightarrow B \quad \vdash s : A}{\vdash ts : B} \rightarrow \text{E}$$

Note that typable terms have unique type. Sometimes we write t^A to emphasize that t is a term of type A .

Definition 2 (Operations with multisets). The letters $M, N, \dots$ stand for multisets of non-negative integers. We write $M \uplus N$ for the *additive* union of multisets, e.g. $\{1, 2, 2\} \uplus \{2, 3, 3\} = \{1, 2, 2, 2, 3, 3\}$. Given a multiset of non-negative integers M and a non-negative integer n , we write $M < n$ if $m < n$ for every $m \in M$. The binary relation $>^1$ between multisets of non-negative integers is defined as follows:

$$M \uplus \{n\} >^1 M \uplus \{m_1, \dots, m_k\} \quad \text{holds for every } n, k, m_1, \dots, m_k \text{ such that } n > m_1, \dots, m_k$$

The *multiset ordering* $M > N$ is defined as the transitive closure of $>^1$.

Theorem 3 (Dershowitz–Manna). *The multiset ordering is well-founded.*

Definition 4. The *degree* $\delta(A)$ of a type A is its height, seen as a tree, that is:

$$\delta(\alpha) \stackrel{\text{def}}{=} 0 \\ \delta(A \rightarrow B) \stackrel{\text{def}}{=} 1 + \max\{\delta(A), \delta(B)\}$$

The *measure* $\#(t)$ of a term is the multiset of degrees of its redexes, that is:

$$\begin{aligned} \#(x^A) &\stackrel{\text{def}}{=} \emptyset \\ \#(\lambda x^A. t) &\stackrel{\text{def}}{=} \#(t) \\ \#(ts) &\stackrel{\text{def}}{=} \#(t) \uplus \#(s) \uplus \begin{cases} \{\delta(A \rightarrow B)\} & \text{if } t \text{ is of the form } t = \lambda x^A. t^B \\ \emptyset & \text{otherwise} \end{cases} \end{aligned}$$

Lemma 5. *Let $\vdash t : A$ and $\vdash s : B$. Suppose that $n \geq 0$ is such that $\#(t) < n$ and $\#(s) < n$. Then $\#(t\{x^B := s\}) < \max\{n, 1 + \delta(B)\}$.*

Proof. By induction on t :

1. Variable, $t = x^B$, with $A = B$: Then $\#(t\{x^B := s\}) = \#(s) < n \leq \max\{n, 1 + \delta(B)\}$.
2. Variable, $t = y^A \neq x$: Then $\#(t\{x^B := s\}) = \#(y) = \emptyset < n \leq \max\{n, 1 + \delta(B)\}$.
3. Abstraction, $t = \lambda y^C. u^D$, with $A = (C \rightarrow D)$: Note that $\#(u) = \#(\lambda y^C. u) < n$ by hypothesis, so $\#(t\{x^B := s\}) = \#(\lambda y^C. u\{x^B := s\}) = \#(u\{x^B := s\}) < \max\{n, 1 + \delta(B)\}$ by IH.

4. Application, $t = t_1^{C \rightarrow A} t_2^C$: Note that $\#(t_1) \subseteq \#(t) < n$ and similarly $\#(t_2) \subseteq \#(t) < n$ by hypothesis, so $\#(t_1) < n$ and $\#(t_2) < n$. By IH:

$$\#(t_1\{x^B := s\}) < \max\{n, 1 + \delta(B)\} \quad \text{and} \quad \#(t_2\{x^B := s\}) < \max\{n, 1 + \delta(B)\}$$

We consider two subcases, depending on whether $t_1\{x^B := s\}$ is an abstraction or not:

- 4.1 If $t_1\{x^B := s\}$ is not an abstraction, it is immediate to conclude, given that:

$$\#(t\{x^B := s\}) = \#(t_1\{x^B := s\}) \uplus \#(t_2\{x^B := s\}) < \max\{n, 1 + \delta(B)\}$$

- 4.2 If $t_1\{x^B := s\}$ is an abstraction, *i.e.* of the form $t_1\{x^B := s\} = \lambda y^C. p^A$: then there are two possibilities, either t_1 is an abstraction, or $t_1 = x$ and s is an abstraction:

- 4.2.1 If $t_1 = \lambda y^C. r^A$, then $\{\delta(C \rightarrow A)\} \subseteq \#(t_1 t_2) = \#(t) < n$ by hypothesis. Hence:

$$\#(t\{x^B := s\}) = \#(t_1\{x^B := s\}) \uplus \#(t_2\{x^B := s\}) \uplus \{\delta(C \rightarrow A)\} < \max\{n, 1 + \delta(B)\}$$

- 4.2.2 If $t_1 = x$ and $s = \lambda y^C. p^A$, where $B = (C \rightarrow A)$, then $\delta(C \rightarrow A) = \delta(B) < 1 + \delta(B)$. Hence:

$$\#(t\{x^B := s\}) = \#(t_1\{x^B := s\}) \uplus \#(t_2\{x^B := s\}) \uplus \{\delta(C \rightarrow A)\} < \max\{n, 1 + \delta(B)\}$$

□

Lemma 6. *Let t be a typable term, and let $t \rightarrow t'$ be a β -step that results from contracting the rightmost redex of maximum degree in t . Then $\#(t) > \#(s)$.*

Proof. By induction on t .

1. Variable, $t = x$: Vacuously true.
2. Abstraction, $t = \lambda x. s$: Immediate by IH.
3. Application, $t = t_1 t_2$. We consider three subcases, depending on whether the step $t \rightarrow t'$ is at the root, internal to t_1 , or internal to t_2 :

- 3.1 Reduction at the root: then $t_1 = \lambda x^A. s^B$ and the step is of the form:

$$t = (\lambda x^A. s) t_2 \rightarrow s\{x^A := t_2\} = t'$$

Note that:

$$\#(t) = \#(s) \uplus \#(t_2) \uplus \{\delta(A \rightarrow B)\}$$

Note that the degree of the contracted redex is $\delta(A \rightarrow B)$ and, since it is the rightmost redex of maximum degree, $\#(s) < \delta(A \rightarrow B)$ and $\#(t_2) < \delta(A \rightarrow B)$. Hence by Lem. 5

$$\#(t') = \#(s\{x^A := t_2\}) < \max\{\delta(A \rightarrow B), 1 + \delta(A)\} = \delta(A \rightarrow B)$$

Therefore $\#(t) > \#(t')$.

- 3.2 Reduction internal to t_1 : then the step is of the form:

$$t = t_1 t_2 \rightarrow t'_1 t_2 = t'$$

where $t_1 \rightarrow t'_1$ again results from contracting the rightmost redex of maximum degree in t_1 . By IH we have that $\#(t_1) > \#(t'_1)$. We consider two subcases, depending on whether t_1 is an abstraction or not:

- 3.2.1 If t_1 is an abstraction, *i.e.* $t_1 = \lambda x^A. p^B$. Then t'_1 is also an abstraction and, by subject reduction, it must be of the form $t'_1 = \lambda x^A. q^B$. Hence:

$$\#(t) = \#(t_1) \uplus \#(t_2) \uplus \{\delta(A \rightarrow B)\} > \#(t'_1) \uplus \#(t_2) \uplus \{\delta(A \rightarrow B)\} = \#(t')$$

3.2.2 If t_1 is not an abstraction, we consider two further subcases, depending on whether t'_1 is an abstraction or not:

3.2.2.1 If t'_1 is an abstraction, i.e. $t'_1 = \lambda x^A. q^B$, then note that, since t_1 is not an abstraction, it must be an application, and the step $t_1 \rightarrow t'_1$ must contract a redex at the root. This means that t_1 is of the form $t_1 = (\lambda y^C. p^{A \rightarrow B}) r$, with $t'_1 = p\{y^C := r\} = \lambda x^A. q^B$. Hence there are two possibilities, either p is an abstraction or $p = y^C$ and r is an abstraction.

- If p is an abstraction, then $p = \lambda z^A. q'^B$, so the step $t \rightarrow t'$ is of the form:

$$(\lambda y^C. \lambda z^A. q') r t_2 \rightarrow (\lambda z^A. q'\{y^C := r\}) t_2$$

Note that the degree of the contracted redex is $\delta(C \rightarrow (A \rightarrow B))$ and, since it is the rightmost redex of maximum degree, $\#(q') < \delta(C \rightarrow (A \rightarrow B))$ and $\#(r) < \delta(C \rightarrow (A \rightarrow B))$. Hence by Lem. 5

$$\#(q'\{y^C := r\}) < \max\{\delta(C \rightarrow (A \rightarrow B)), 1 + \delta(C)\} = \delta(C \rightarrow (A \rightarrow B))$$

Therefore:

$$\begin{aligned} \#(t) &= \#(q') \uplus \#(r) \uplus \#(t_2) \uplus \{\delta(C \rightarrow (A \rightarrow B))\} \\ &> \#(q'\{y^C := r\}) \uplus \#(t_2) \uplus \{\delta(A \rightarrow B)\} \\ &= \#(t') \end{aligned}$$

- If $p = y^C$ and $r = \lambda x^A. q^B$ then $C = (A \rightarrow B)$ and the step $t \rightarrow t'$ is of the form:

$$t = (\lambda y^{A \rightarrow B}. y^{A \rightarrow B}) (\lambda x^A. q^B) t_2 \rightarrow (\lambda x^A. q^B) t_2 = t'$$

Then:

$$\#(t) = \#(q^B) \uplus \#(t_2) \uplus \{\delta((A \rightarrow B) \rightarrow A \rightarrow B)\} > \#(q^B) \uplus \#(t_2) \uplus \{\delta(A \rightarrow B)\} = \#(t')$$

3.2.2.2 If t'_1 is not an abstraction, then it is immediate to conclude, as:

$$\#(t) = \#(t_1) \cup \#(t_2) > \#(t'_1) \cup \#(t_2) = \#(t')$$

3.3 Reduction internal to t_2 : then the step is of the form:

$$t = t_1 t_2 \rightarrow t_1 t'_2 = t'$$

where $t_2 \rightarrow t'_2$ again results from contracting the rightmost redex of maximum degree in t_2 . By IH we have that $\#(t_2) > \#(t'_2)$. From this we conclude that $\#(t_1 t_2) > \#(t_1 t'_2)$, as required.

□

Theorem 7. *The simply typed λ -calculus is weakly normalizing.*

Proof. An easy corollary of Thm. 3 and Lem. 6.

□